

Install docker on EC2 and explore the docker commands (docker images, containers, volumes, network)

Docker installation :

```
ubuntu@ip-172-31-21-29:~$ sudo apt install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  docker-ce-rootless-extras libslirp0 pigz slirp4netns
Suggested packages:
  cgroupfs-mount | cgroup-lite docker-model-plugin
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin libslirp0 pigz slirp4netns
0 upgraded, 9 newly installed, 0 to remove and 68 not upgraded.
Need to get 91.3 MB of archives.
After this operation, 364 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://us-west-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://us-west-2.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libslirp0 amd64 4.7.0-1ubuntu3 [63.8 kB]
Get:3 http://us-west-2.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 slirp4netns amd64 1.2.1-1build2 [34.9 kB]
Get:4 https://download.docker.com/linux/ubuntu noble/stable amd64 containerd.io amd64 2.2.1-1-ubuntu.24.04-noble [23.4 MB]
Get:5 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-cli amd64 5:29.1.3-1-ubuntu.24.04-noble [16.3 MB]
Get:6 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce amd64 5:29.1.3-1-ubuntu.24.04-noble [21.0 MB]
Get:7 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-buildx-plugin amd64 0.30.1-1-ubuntu.24.04-noble [16.4 MB]
Get:8 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-ce-rootless-extras amd64 5:29.1.3-1-ubuntu.24.04-noble [6383 kB]
Get:9 https://download.docker.com/linux/ubuntu noble/stable amd64 docker-compose-plugin amd64 5.0.1-1-ubuntu.24.04-noble [7713 kB]
Fetched 91.3 MB in 1s (86.3 MB/s)
Selecting previously unselected package containerd.io.
(Reading database ... 71735 files and directories currently installed.)
Preparing to unpack .../0-containerd.io_2.2.1-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking containerd.io (2.2.1-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce-cli.
Preparing to unpack .../1-docker-ce-cli_5%3a29.1.3-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce-cli (5:29.1.3-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce.
Preparing to unpack .../2-docker-ce_5%3a29.1.3-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce (5:29.1.3-1-ubuntu.24.04-noble) ...
Selecting previously unselected package pigz.
Preparing to unpack .../3-pigz_2.8-1_amd64.deb ...
Unpacking pigz (2.8-1) ...
Selecting previously unselected package docker-buildx-plugin.
Preparing to unpack .../4-docker-buildx-plugin_0.30.1-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-buildx-plugin (0.30.1-1-ubuntu.24.04-noble) ...
Selecting previously unselected package docker-ce-rootless-extras.
Preparing to unpack .../5-docker-ce-rootless-extras_5%3a29.1.3-1-ubuntu.24.04-noble_amd64.deb ...
Unpacking docker-ce-rootless-extras (5:29.1.3-1-ubuntu.24.04-noble) ...
```

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-21-29:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-01-06 06:15:46 UTC; 15s ago
 TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 2391 (dockerd)
     Tasks: 9
    Memory: 84.2M (peak: 84.7M)
       CPU: 366ms
    CGroup: /system.slice/docker.service
            └─2391 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock
```

## Configured mysql container using bind mount :

```
ubuntu@ip-172-31-21-29:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
ubuntu@ip-172-31-21-29:~$ mkdir -p cont_data/sql_server01
ubuntu@ip-172-31-21-29:~$ docker run --name sqlcont01 \
> -v /home/ubuntu/cont_data/sql_server01:/var/lib/mysql \
man-db/ misc/
> -v /home/ubuntu/cont_data/sql_server01:/var/lib/mysql \
> -e MYSQL_ROOT_PASSWORD=test123 \
> -d mysql:8.0.44-debian
docker: invalid reference format

Run 'docker run --help' for more information
ubuntu@ip-172-31-21-29:~$ docker run --name sqlcont01-v /home/ubuntu/cont_data/sql_server01:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=test123 -d mysql:8.0.44-debian
docker: invalid reference format

Run 'docker run --help' for more information
ubuntu@ip-172-31-21-29:~$ docker run \
--name sqlcont01-v \
-v /home/ubuntu/cont_data/sql_server01:/var/lib/mysql \
-e MYSQL_ROOT_PASSWORD=test123 \
-d mysql:8.0.44-debian
Unable to find image 'mysql:8.0.44-debian' locally
8.0.44-debian: Pulling from library/mysql
c9aa0d54cd11: Pull complete
0347dcb76707: Pull complete
b48a28f3e210: Pull complete
cc0a8526c52a: Pull complete
e2c675c73718: Pull complete
d69cd916644c: Pull complete
6c3a88f9c40f: Pull complete
b299e24801f4: Pull complete
c1c3a90b4073: Pull complete
dabe1e9c791f: Pull complete
d876c61fc2a6: Pull complete
d09e4c81218d: Pull complete
f50a6e29ca8f: Download complete
6b8bc3f59023: Download complete
Digest: sha256:91f968e434fa4c9ea47e3fac55f83da86b080f903bbce1201931e97f4194bd5e
Status: Downloaded newer image for mysql:8.0.44-debian
42e2cb9fd9b2d73489ac964f7027e232e4db881ec924765eb1bdf22d3bbddb9
ubuntu@ip-172-31-21-29:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
42e2cb9fd9b2b  mysql:8.0.44-debian  "docker-entrypoint.s..."  7 seconds ago  Up 7 seconds  3306/tcp, 33060/tcp  sqlcont01-v
```

## Configured mysql container using volume mount

```
ubuntu@ip-172-31-21-29:~$ docker create volume sql02
Unable to find image 'volume:latest' locally
Error response from daemon: pull access denied for volume, repository does not exist or may require 'docker login'
ubuntu@ip-172-31-21-29:~$ docker create volume sql03
Unable to find image 'volume:latest' locally
Error response from daemon: pull access denied for volume, repository does not exist or may require 'docker login'
ubuntu@ip-172-31-21-29:~$ docker volume ls
DRIVER      VOLUME NAME
ubuntu@ip-172-31-21-29:~$ docker volume create mysql_data
mysql_data
ubuntu@ip-172-31-21-29:~$ docker run --name sql02 \
> -v sql02:/var/lib/mysql \
> -e MYSQL_ROOT_PASSWORD=test123 \
> -d mysql:8.0.44-debian
08236804f1031b3268759fc022e5836e39e77446a3d29b54fdf9eb36b70a1685
ubuntu@ip-172-31-21-29:~$
```